

User Manual - Strategic Board Game

How to Launch the Program

Compilation

To compile the program, you must have the « jeu_rafael_thomas_td2_groupesae31.c », « board.h », « board.o » and « main_example.c » files in the same folder and use the following command in your terminal:

```
gcc -o game jeu_rafael_thomas_td2_groupesae31.c board.o
```

Execution

To start the game, run:

```
./game
```

Game Overview

This is a two-player strategic board game played on a 6x6 grid with two goal zones (North and South). Players compete to be the first to place a piece in the opponent's goal zone.

Game Rules

Setup Phase

- *The game uses 12 pieces total: 4 pieces of size 1, 4 of size 2, and 4 of size 3*
- *Players alternate placing pieces on their starting line (South player on row 1, North player on row 6)*
- *Each player places 2 pieces of each size*

Game Phase

Turn Structure:

1. *Select a piece from your closest occupied row*
2. *Move the piece a number of steps equal to its size*
3. *Each step must be in a cardinal direction (North, South, East, West)*

Movement Rules:

- *All steps except the last must land on empty squares*
- *The last step can land on an occupied square or the opponent's goal*
- *If landing on an occupied square, your piece continues moving with steps equal to the size of the piece you landed on*
- *You can't enter the same square from the same direction twice in one turn*

Special Move - Swap: *When your piece lands on another piece, you can choose to:*

- **Bounce:** *Continue moving*
- **Swap:** *Exchange positions and place the displaced piece anywhere on the board*

Winning: *The first player to move a piece into the opponent's goal zone wins!*

Using the Interface

During Placement Phase

- *Enter the piece size (1-3)*
- *Enter the column number (1-6)*

During Game Phase

1. **Select a piece:** *Enter row and column (1-6)*
2. **Move the piece:** *Use direction commands:*
 - *n = North*
 - *s = South*
 - *e = East (right)*
 - *o = West (left)*
 - *g = Goal (to enter the opponent's goal zone)*

Special Commands

- *u = Undo last step*
- *c = Cancel entire movement (generally to avoid bugs)*
- *b = Bounce (continue movement when landing on a piece)*
- *s = Swap (exchange with the piece you landed on)*

Good luck and enjoy the game!